

🔥 Module A: Tabular Solution of the Simplex Method

Setup. For a standard-form LP

$$\min c^\top x \quad \text{s.t.} \quad Ax = b, \quad x \geq 0,$$

simplex moves from one **BFS** to another by changing the basis one column at a time.

Convert to Canonical Form After row operations that make the current basis matrix equal to the identity,

$$[I_m \quad Y \quad y_0] \implies x_B = y_0, x_N = 0.$$

Hence:

- if $y_0 \geq 0$, the current basic solution is a **BFS**;
- the current basis columns are exactly the columns that form I_m .

For a **minimization** problem:

- if all reduced costs are ≥ 0 , current BFS is optimal;
- if some $r_q < 0$, increasing x_q can decrease the objective.

Entering / leaving / pivot. Choose an entering variable x_q with $r_q < 0$ (often the most negative one). Let the corresponding nonbasic column in the current tableau be $y_{\cdot q}$. To keep feasibility, use the ratio:

$$p = \arg \min_i \underbrace{\left\{ \frac{(y_0)_i}{y_{iq}} : y_{iq} > 0 \right\}}_{\text{MAX dist can go under } x \geq 0}.$$

Reduced costs. If $A = [B \ D]$ and $c^\top = [c_B^\top \ c_N^\top]$, then

$$r_N^\top = \underbrace{c_N^\top}_{\text{cost increase by increasing } x_N} - \underbrace{c_B^\top B^{-1} D}_{\text{cost decrease by compensation}}.$$

In the tableau, the bottom row under the nonbasic columns is the reduced-cost row.

Then:

- x_q enters the basis;
- the basic variable in row p leaves;
- pivot on (p, q) , i.e. use row operations so the new basis columns become I_m again.

📌 Practice: Textbook 16.4 (tabular simplex)

Consider the linear program

$$\text{maximize } 2x_1 + x_2, \quad \text{subject to } 0 \leq x_1 \leq 5, \quad 0 \leq x_2 \leq 7, \quad x_1 + x_2 \leq 9.$$

- Convert the problem to standard form using slack variables.
- Starting from the slack-variable basis, solve the LP by the **tableau simplex method**.
- State the optimal solution and optimal objective value for the original LP.

Answer: 1. **Standard Form:** To solve the maximization problem using the standard tabular simplex, we convert it to a minimization problem: $\min -2x_1 - x_2$. We add slack variables $x_3, x_4, x_5 \geq 0$ to convert inequalities to equalities:

$$x_1 + x_3 = 5, \quad x_2 + x_4 = 7, \quad x_1 + x_2 + x_5 = 9.$$

2. **Tabular Simplex Iterations:** *Initial Tableau* (Basis: x_3, x_4, x_5):

Basic	x_1	x_2	x_3	x_4	x_5	RHS
x_3	1	0	1	0	0	5
x_4	0	1	0	1	0	7
x_5	1	1	0	0	1	9
$r^\top$	-2	-1	0	0	0	0

Entering variable: x_1 (most negative reduced cost $r_1 = -2$).
Ratio test: $\min\{\frac{5}{1}, -\frac{9}{1}\} = 5$. Leaving variable: x_3 . Pivot at row 1, col 1.
Iteration 1 (Basis: x_1, x_4, x_5):

Basic	x_1	x_2	x_3	x_4	x_5	RHS
x_1	1	0	1	0	0	5
x_4	0	1	0	1	0	7
x_5	0	1	-1	0	1	4
$r^\top$	0	-1	2	0	0	10

Entering variable: x_2 ($r_2 = -1$).
Ratio test: $\min\{-\frac{7}{1}, \frac{4}{1}\} = 4$. Leaving variable: x_5 . Pivot at row 3, col 2.
Iteration 2 (Basis: x_1, x_4, x_2):

Basic	x_1	x_2	x_3	x_4	x_5	RHS
x_1	1	0	1	0	0	5
x_4	0	0	1	1	-1	3
x_2	0	1	-1	0	1	4
$r^\top$	0	0	1	0	1	14

All reduced costs are ≥ 0 , so the current BFS is optimal.

3. **Optimal Solution:** The optimal solution is $(x_1^*, x_2^*) = (5, 4)$ with slacks $(x_3, x_4, x_5) = (0, 3, 0)$. The RHS of the cost row indicates the negative of the min objective $(-(-14))$. Thus, the optimal objective value for the original maximization problem is 14.

Standard primal/dual pair (lecture form). For

$$\min c^\top x \quad \text{s.t.} \quad Ax = b, \quad x \geq 0,$$

the dual is

$$\max \lambda^\top b \quad \text{s.t.} \quad \lambda^\top A \leq c^\top.$$

Min-max intuition. The primal can be written as

$$\min_{x \geq 0} \max_{\lambda} (c^\top x - \lambda^\top (Ax - b)).$$

If $Ax \neq b$, the inner maximization can blow up, so any primal optimum must satisfy $Ax = b$.

Swapping the order gives

$$\max_{\lambda} \min_{x \geq 0} (c^\top x - \lambda^\top (Ax - b)).$$

For fixed λ , the inner minimum is finite only if

$$c^\top - \lambda^\top A \geq 0,$$

which yields the dual constraints.

Useful max- $\leq$ form. If the primal is

$$\max c^\top x \quad \text{s.t.} \quad Ax \leq b, \quad x \geq 0,$$

then the dual is

$$\min b^\top \lambda \quad \text{s.t.} \quad A^\top \lambda \geq c, \quad \lambda \geq 0.$$

Weak duality. For any primal-feasible x and dual-feasible y ,

$$c^\top x \leq b^\top y$$

That is:

- a dual-feasible point gives an **upper bound** on a primal maximization problem;
- a primal-feasible point gives a **lower bound** on a dual minimization problem.

Strong duality / Duality Theorem If the primal problem has an optimal solution, then so does the dual, and the optimal values of their respective objective functions are equal.

Practice: Textbook 17.3 (duality)

Consider the linear program

$$\text{maximize } 2x_1 + 3x_2$$

subject to

$$x_1 + 2x_2 \leq 4, \quad 2x_1 + x_2 \leq 5, \quad x_1, x_2 \geq 0.$$

1. Solve the primal problem using tabular simplex.
2. Write down the dual problem.
3. Solve the dual and verify that the two optimal objective values coincide.

Answer: 1. **Primal Tabular Simplex:** Convert to standard minimization form ($\min -2x_1 - 3x_2$) with slack variables $x_3, x_4 \geq 0$:

Basic	x_1	x_2	x_3	x_4	RHS
x_3	1	2	1	0	4
x_4	2	1	0	1	5
$r^\top$	-2	-3	0	0	0

Entering: x_2 ($r_2 = -3$). Ratio test: $\min\{\frac{4}{2}, \frac{5}{1}\} = 2$. Leaving: x_3 . Pivot at (1, 2).

Iteration 1 (Basis: x_2, x_4):

Basic	x_1	x_2	x_3	x_4	RHS
x_2	1/2	1	1/2	0	2
x_4	3/2	0	-1/2	1	3
$r^\top$	-1/2	0	3/2	0	6

Entering: x_1 ($r_1 = -1/2$). Ratio test: $\min\{\frac{2}{1/2}, \frac{3}{3/2}\} = \min\{4, 2\} = 2$. Leaving: x_4 . Pivot at (2, 1).

Iteration 2 (Basis: x_2, x_1):

Basic	x_1	x_2	x_3	x_4	RHS
x_2	0	1	2/3	-1/3	1
x_1	1	0	-1/3	2/3	2
$r^\top$	0	0	4/3	1/3	7

Optimal! Primal solution: $(x_1^*, x_2^*) = (2, 1)$, maximum objective value = 7.

2. **The Dual Problem:** Let $y_1, y_2 \geq 0$ be the dual variables for the two inequality constraints. The dual is:

$$\min 4y_1 + 5y_2 \quad \text{s.t.} \quad y_1 + 2y_2 \geq 2, \quad 2y_1 + y_2 \geq 3, \quad y_1, y_2 \geq 0.$$

3. **Solving the Dual via the Primal Tableau:** By Strong Duality, the optimal dual variables (y_1^*, y_2^*) can be read directly from the reduced cost row of the optimal primal tableau, specifically under the primal slack variables x_3 and x_4 . From our final tableau, $r_3 = 4/3$ and $r_4 = 1/3$. Thus, the optimal dual solution is $y_1^* = 4/3$ and $y_2^* = 1/3$.

To verify, we check the dual objective value: $4(\frac{4}{3}) + 5(\frac{1}{3}) = \frac{21}{3} = 7$. Since Primal OPT = 7 = Dual OPT, strong duality is fully verified!